

Incident Response

Condensed Study Notes

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WEKK 5 INCIDENCE RESPONSE

This section introduces the concept of 'incidence response', which pertains to how systems react when an event or 'incidence' occurs.

Key terms:

- **Incidence:** An event that occurs within a system.
- **Response:** The action or reaction taken by a system when an incidence occurs.

The core idea is the relationship between an event and the system's subsequent behavior.

Introduction

Incident response is a structured, strategic method for dealing with cyber attacks. Its goal is to lessen the impact of attacks, reduce the time it takes to recover, and lower overall costs.

Incident response is a part of incident management. Incident management is the broader company-wide process for handling cyber attacks, involving many departments like executive, legal, HR, communications, and IT. Incident response specifically focuses on the technical cybersecurity aspects and actions.

Types of security incidents

Understanding security incidents is key to developing effective incident response strategies. The terms relate as follows:

- **Vulnerability:** A weakness in an IT or business system.
- **Threat:** An entity (e.g., hacker, insider) that aims to exploit a vulnerability.
- **Incident:** An attack that succeeds in accessing enterprise resources or putting them at risk.
- **Data Breach:** An incident where attackers successfully compromise sensitive information.

Proactive measures, like fixing vulnerabilities and preparing for common incidents, are crucial. Common types of security incidents include:

- Privilege escalation attacks
- Man-in-the-middle attacks
- Malware attacks
- Unauthorized access attempts
- Advanced persistent threats (APTs)
- Web application attacks
- Insider threats

- Phishing attacks
- Password attacks
- Denial-of-service (DoS) attacks

How to create an incident response plan

Creating a robust incident response plan involves proactive preparation before an incident occurs.

Key components for a successful plan include:

1. **Establish a Policy:** A high-level, adaptable document outlining general priorities for handling incidents.
2. **Build an Incident Response Team:** Assemble a team and clearly define roles, responsibilities, and ensure adequate training for all members.
3. **Create Playbooks:** These are detailed, step-by-step guides for specific incident scenarios. Playbooks ensure consistency, efficiency, and effectiveness in both response and training.
4. **Create a Communication Plan:** A vital element for coordinating with various stakeholders, including internal teams (response, executive, legal, HR), customers, partners, law enforcement, and the public.

What is an incident response plan

An incident response plan is an organization's core documentation that guides actions during a security incident. It acts as a detailed map, directing responders from initial detection and assessment through containment and resolution.

The plan specifically details:

- **Who:** Identifies individuals responsible for tasks and provides contact information.
- **What:** Defines which situations constitute actionable security incidents and outlines the appropriate responses.
- **How:** Explains the specific methods team members should use to complete their assigned tasks.
- **When:** Clarifies the circumstances under which specific tasks should be performed.

Incident response frameworks: Phases of incident response

Established incident response frameworks provide high-level guidance for handling security incidents. While specific frameworks (NIST, SANS, ISO, ISACA) vary, they generally describe similar phases:

1. **Preparation/Planning:** Build an incident response team and create policies, processes, and playbooks. Deploy supporting tools and services.
2. **Detection/Identification:** Use IT monitoring to detect, evaluate, validate, and triage security incidents.
3. **Containment:** Take steps to stop an incident from worsening and regain control of IT resources.
4. **Eradication:** Eliminate threat activity (e.g., malware, malicious accounts) and identify exploited vulnerabilities.

5. **Recovery:** Restore normal operations and mitigate identified vulnerabilities.
6. **Lessons Learned:** Review the incident to understand what, when, and how it happened. Identify suboptimal security controls, policies, and procedures, and plan improvements. Update the incident response plan based on these findings.

Types of incident response teams

Several types of incident response teams exist, with three being most common:

- **Computer Incident Response Team (CIRT)**
- **Computer Security Incident Response Team (CSIRT)**
- **Computer Emergency Response Team (CERT)**

Why is an Incident Response Plan Important

An Incident Response Plan (IRP) is crucial for organizations as it provides a structured approach to handling cyber incidents.

Key benefits include:

- **Minimizing Impact:** Enables quick and efficient action, reducing downtime, financial losses, and reputational damage.
- **Improving Detection:** Incorporates continuous monitoring and system reviews to detect threats earlier and prevent breaches.
- **Streamlining Communication:** Ensures all stakeholders (IT, management, employees) are aligned, reducing confusion and speeding up recovery.
- **Building Resilience:** Fosters security awareness among employees, making the organization more robust against attacks.
- **Legal & Regulatory Compliance:** Helps demonstrate commitment to data protection and cybersecurity regulations, avoiding penalties.